Zihao Wang

404-952-0911 | ziw140@ucsd.edu | gdfwj.github.io | github.com/gdfwj

Education

University of California, San Diego La Jolla, CA
Master of Science in Computer Science *Sep. 2024 – Now*

- GPA: 4.0/4.0
- Key courses: Systems for LLMs and AI Agents, Differentiable Programming (A+), ML: Learning Algorithms (A+)

Beihang University Beijing, China
Bachelor of Engineering in Computer Science (Minor in Mathematics and Applied Mathematics) *Sep. 2020 – Jun. 2024*

- GPA: 3.92/4.0
- Key courses: Computer Organization (96), Operating Systems (95), Object Oriented Design Construction (98), Compiler Technology (95), Signal and System (98), Computer Network Experiments (97)

Research Interests

My research interests lie in Machine Learning, Systems for AI, IoT (Internet of Things), and edge computing. I aim to build multi-modal and general AI systems that can assist people across all aspects of daily life. I am currently developing an efficient edge LLM agent system that retrieves and reasons over database and sensor data to answer natural-language queries in SeeLab, advised by Prof. **Tajana Šimunić Rosing** and **Ye Tian**.

Scholarships & Awards

- **2021, 2022, 2023** Academic Merit Scholarship (Awarded to undergraduate students for excelling GPA throughout the school year)
- **2022, 2023** Discipline Competition Scholarship (Awarded to undergraduate students who achieved outstanding rankings in national or university-level discipline competitions)

Publications

1. Ye Tian*, **Zihao Wang***, Onat Gungor, Xiaoran Fan, Tajana Rosing. "MultiLifeQA: A Multi-Domain Health Reasoning Benchmark." *ICLR*, 2026 (under review)
2. Ye Tian, Xiaoyuan Ren, **Zihao Wang**, Onat Gungor, Xiaofan Yu, Tajana Rosing. "DailyLLM: Context-Aware Activity Log Generation Using Multi-Modal Sensors and LLMs." *22nd IEEE International Conference on Mobile Ad-Hoc and Smart Systems (MASS)*, 2025

Research Experience

Efficient Edge–Client Collaborative Agents for Question Answering Oct, 2025–Present
Research Assistant, Advisors: Ye Tian and Tajana Rosing
System Energy Efficiency Lab, University of California, San Diego

- Building an efficient edge–on-prem collaborative agent for MultiLifeQA-style health QA over private databases; implemented initial tools based on smolagents and set up evaluation scaffolding.
- Improving answer accuracy via multi-step reasoning and more reliable tool use.
- Increasing efficiency by optimizing tool prompts and history, as well as segregating on-client LLM inference from on-edge tool calls.
- Designing a privacy-preserving edge–client PC pipeline: executing data access and SQL on-device at the edge, running LLM inference locally on the user's PC, keeping sensitive data off-cloud.
- Targeting submission to **ACL 2026**.

Multidimensional Health QA Benchmark (MultiLifeQA) Jul, 2025–Sep, 2025
Research Assistant, Advisors: Ye Tian and Tajana Rosing
System Energy Efficiency Lab, University of California, San Diego

- Proposed a large-scale QA benchmark comprising 22,573 questions for multidimensional health reasoning using database-stored lifestyle data.
- Designed a framework that transforms structured datasets (e.g., Excel files) into a MySQL database and automatically generates QA pairs from question templates and database entries.
- Developed context-based and database-augmented evaluation protocols with fine-grained metrics for validity, execution quality, and answer accuracy.
- Led experiments on eight open-source and three proprietary LLMs to analyze their reasoning capabilities and limitations in multidimensional health understanding, and released the full dataset and benchmark at https://github.com/gdfwj/MultilifeQA.
- Paper submitted to **ICLR 2026** (under review).

DailyLLM: Multimodal Activity Log Generation Jan, 2025–April, 2025
Research Assistant, Advisors: Ye Tian and Tajana Rosing
System Energy Efficiency Lab, University of California, San Diego

- Proposed **DailyLLM**, a lightweight LLM-based system for context-aware activity log generation and summarization using multimodal smartphone and smartwatch sensors.
- Designed and generated the IMU-based dataset for activity recognition, including data preprocessing, labeling, and augmentation.
- Implemented LLM fine-tuning and evaluation pipelines for all four tasks (including activity prediction from IMU, scene understanding from audio, log generation, and summarization) and conducted controlled comparisons against baselines.
- Developed the **project website** for dataset and code release.
- Paper accepted to **The 22nd IEEE International Conference on Mobile Ad-Hoc and Smart Systems (MASS 2025)**.

Reconstruction of perceived face images from fMRI Apr, 2023–Dec, 2023
Research Assistant, Advisors: Hui Zhang
Medical-Engineering Innovation Research Institute

- Designed and trained a model to reconstruct the face image perceived in brain from fMRI data.
- Implemented a pre-training strategy on the CelebA dataset, utilizing the features extracted by VGGFace as conditions, as a solution to the scarcity of fMRI-face paired data
- Designed a double-flow architecture of the discriminator for GAN to challenge the generator.
- Manuscript was published on arXiv.

Select Course Projects

Compiler Design and Implementation Sep. 2022 – Dec. 2022
- Implemented a SysY-to-MIPS compiler covering lexical and grammatical analysis, IR generation, object code generation, and classic optimizations; realized full functionality.
- Gained experience structuring a multi-file codebase with complex logic.

Computer Organization Project: Five-Stage Pipelined MIPS CPU Sep. 2021 – Jan. 2022
- Implemented a five-stage pipelined MIPS CPU in Verilog with interrupt/exception handling and simple I/O; built a single-cycle baseline then upgraded to a pipelined design.
- Strengthened understanding of CPU microarchitecture and hardware coding.

Operating Systems Project: MIPS-based OS Mar. 2022 – Jun. 2022
- Built a simple OS from scratch on MIPS: boot, printf, memory management, process management, user/kernel modes, system calls, file system, and shell.
- Implemented basic threading management and synchronization system calls.